

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL AS CHEMISTRY (9620)

Unit 2: Organic 1 and Physical 1

Monday 20 January 2020 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- the Periodic Table/Data Sheet, provided as an insert
- a ruler with millimetre measurements
- a scientific calculator, which you are expected to use where appropriate.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do **not** write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- All working must be shown.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 70.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	



Answer **all** questions in the spaces provided.

Do not write
outside the
box

0 1

Hydrocarbon **A** is obtained by the cracking of an alkane.

0 1 . 1

State what is meant by the term cracking.

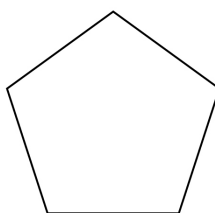
Give **one** reason why some alkanes are cracked.

[2 marks]

Meaning of the term cracking _____

Reason for cracking _____

The skeletal formula of **A** is shown.



0 1 . 2

Draw the skeletal formula of an isomer of **A** that is **not** a cyclic alkane.

[1 mark]



0 1 . 3

Identify a reagent that can be used in a simple test-tube reaction to distinguish between **A** and an isomer of **A** that is **not** a cyclic alkane.

State what is observed.

[3 marks]

Reagent _____

Observation with **A** _____

Observation with an isomer of **A** that is **not** a cyclic alkane _____

0 1 . 4

A cracking reaction forms **A**, ethane and ethene in a 1:1:2 ratio.

Write an equation for this reaction.

[1 mark]

7

Turn over for the next question

Turn over ►



0	2
---	---

This question is about some reactions of propene.

Propene reacts with hydrogen bromide to form two different products.

0	2	.	1
---	---	---	---

Name the mechanism for this reaction.

[1 mark]

0	2	.	2
---	---	---	---

Draw the skeletal formula of the major product formed in the reaction of propene with hydrogen bromide.

Explain why this is the major product.

[3 marks]

Skeletal formula

Explanation

0	2	.	3
---	---	---	---

Propene can form poly(propene).

Explain why propene is reactive and poly(propene) is **not** reactive.

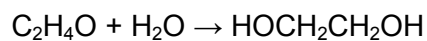
[2 marks]

6



0	3
---	---

In **acidic** conditions, epoxyethane reacts with water to form ethane-1,2-diol.



0	3	.	1
---	---	---	---

Outline the mechanism for this reaction.

Give **one** use of ethane-1,2-diol.

[4 marks]

Mechanism

Use _____

0	3	.	2
---	---	---	---

A sample of 4.3 g of epoxyethane reacts with an excess of water to form 4.6 cm³ of ethane-1,2-diol.

Calculate the percentage yield for the formation of ethane-1,2-diol.

The density of ethane-1,2-diol is 1.11 g cm⁻³

[4 marks]

Percentage yield _____

8

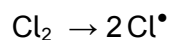
Turn over ►



0	4
---	---

This question is about halogenoalkanes.

The first step in the mechanism of the reaction between chlorine and methane to form chloromethane is shown.



0	4	.	1
---	---	---	---

Name the mechanism of the reaction to form chloromethane from methane.

[1 mark]

0	4	.	2
---	---	---	---

Give equations for two steps in this reaction that involve $\text{Cl}^\bullet$ as a reactant.

[2 marks]

1

2

0	4	.	3
---	---	---	---

Chloromethane reacts with an excess of ammonia.

Outline a mechanism for this reaction.

[4 marks]

0	4	.	4
---	---	---	---

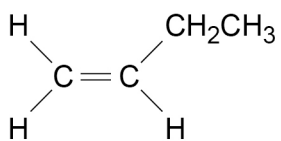
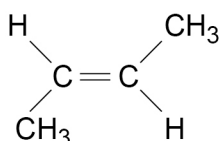
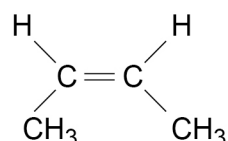
Cyanide acts as a nucleophile in the reaction between potassium cyanide and chloromethane.

State why cyanide is able to act as a nucleophile in this reaction.

[1 mark]



Hydroxide ions react with 2-chlorobutane to form the three isomers shown.

**P****Q****R**

0 4 . 5 State the role of hydroxide ions in this reaction.

[1 mark]

0 4 . 6 **P** and **Q** are isomers.

Name this type of isomerism.

[1 mark]

0 4 . 7 **Q** and **R** are stereoisomers.

Define the term stereoisomers.

[2 marks]

0 4 . 8 **R** is Z-but-2-ene.

State why **R** is the Z isomer.

[1 mark]



0 5

This question is about alcohols and their oxidation products.

0 5 . 1

Some alcohols can be oxidised.

Identify a suitable oxidising agent.

State the colour change observed in the oxidation.

[2 marks]

Oxidising agent _____

Colour change _____

0 5 . 2

Butan-1-ol is oxidised to form an aldehyde.

Write an equation for this reaction.

You should use [O] to represent the oxidising agent.

State how the aldehyde is obtained from the reaction mixture.

[2 marks]

Equation

How aldehyde is obtained

0 5 . 3

Draw the structure of the carboxylic acid formed by the oxidation of butan-1-ol.

[1 mark]

0 5 . 4

Use IUPAC rules to name the organic compound formed when butan-2-ol is oxidised.

[1 mark]

0 5 . 5

Compounds **S** and **T** are isomers with $M_r = 58.0$ When heated with Tollens' reagent, **S** forms a silver mirror but **T** does not.Draw the structures of **S** and **T**.

[2 marks]

S**T**

0 5 . 6

Draw the structure of the alcohol with molecular formula $C_5H_{12}O$ that is **not** easily oxidised.

[1 mark]

Question 5 continues on the next page

Turn over ►



0 5 . 7

Compound **U** has $M_r = 60.0$ The infrared spectrum of **U** has absorptions in the ranges $2500\text{--}3000\text{ cm}^{-1}$ and $1680\text{--}1750\text{ cm}^{-1}$ Name compound **U**.Identify a reagent that can be used in a simple test-tube reaction to confirm the presence of the functional group in **U**.

State what is observed.

[3 marks]Name of **U** _____

Reagent _____

Observation _____

0 5 . 8

Which compound is a secondary alcohol?

Tick (✓) **one** box.**[1 mark]**

2,2-dimethylpropan-1-ol

☐

2,3-dimethylbutan-2-ol

☐

3,3-dimethylpentan-2-ol

☐

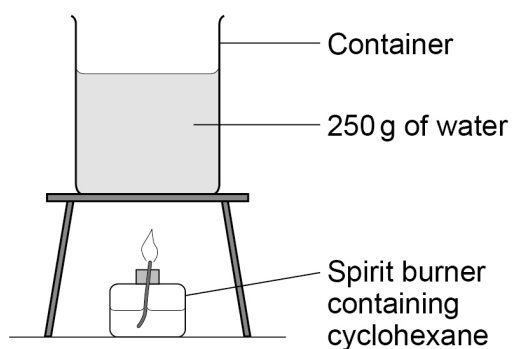
13

0 6

This question is about cyclohexane, C_6H_{12}

0 6 . 1

A student collects data to calculate the enthalpy of combustion of liquid cyclohexane.



Method

- Weigh the spirit burner containing the cyclohexane and place it under the container of water.
- Measure the initial temperature of the water.
- Allow the cyclohexane to burn for five minutes.
- Measure the maximum temperature of the water.
- Reweigh the spirit burner containing the cyclohexane.

Results

Initial temperature of water	18.5 °C
Maximum temperature of water	49.0 °C
Mass of cyclohexane burned	1.51 g

Calculate a value for the enthalpy of combustion, in kJ mol^{-1} , of cyclohexane.

The specific heat capacity of water, $c = 4.18 \text{ J K}^{-1} \text{ g}^{-1}$
 You should ignore the heat capacity of the container.

[4 marks]Enthalpy of combustion _____ kJ mol^{-1} **Turn over ►**

A value for the enthalpy of combustion of cyclohexane can be calculated from mean bond enthalpy data.

0 6 . 2

Define the term mean bond enthalpy.

[1 mark]

0 6 . 3

Cyclohexane (C_6H_{12}) burns in oxygen as shown.

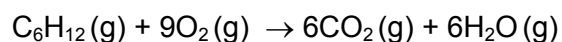


Table 1 shows some mean bond enthalpy data.

Table 1

	C—C	C—H	O=O	C=O	O—H
Mean bond enthalpy / kJ mol^{-1}	348	412	496	743	463

Use the equation and the data in **Table 1** to calculate a value for the enthalpy of combustion, in kJ mol^{-1} , of cyclohexane.

[3 marks]

Enthalpy of combustion _____ kJ mol^{-1}



06.4

The enthalpy of combustion value calculated using bond enthalpy data in **Question 06.3** is different from the value calculated in the experiment in **Question 06.1**.

Give **two** reasons why these values are different.

[2 marks]

Reason 1 _____

Reason 2 _____

10

Turn over for the next question

Turn over ►



0 7Compound **Y** was analysed in a combustion experiment.**0 7 . 1**Complete combustion of 0.220 g of **Y** formed 0.524 g of carbon dioxide and 0.267 g of water.Calculate the percentage by mass of carbon and of hydrogen in **Y**.**[3 marks]**

C _____ %

H _____ %

0 7 . 2Compound **Y** contains carbon, hydrogen and oxygen only.Use your answers to **Question 07.1** to calculate the empirical formula of **Y**.(If you were unable to answer **Question 07.1** you should use the values 68.1% carbon and 13.7% hydrogen. These are **not** the correct values.)**[3 marks]**

Empirical formula _____



07.3

Compound **Y** and compound **Z** both have an absorption at 3330 cm^{-1} in their infrared spectra.

Z is a symmetrical molecule with molecular formula $\text{C}_3\text{H}_8\text{O}$

Draw the displayed formula of **Z**

[1 mark]

7

Turn over for the next question

Turn over ►



0	8
---	---

The polymer poly(tetrafluoroethene) is made from the monomer C_2F_4

0	8	.	1
---	---	---	---

Draw the repeating unit of poly(tetrafluoroethene).

[1 mark]

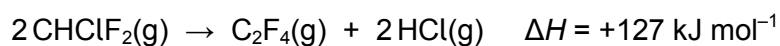
0	8	.	2
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Define the term standard enthalpy of formation.

[2 marks]



0 8 . 3 The monomer C_2F_4 can be made as shown in the equation.



Some standard enthalpies of formation are shown.

Substance	$C_2F_4(g)$	$HCl(g)$
$\Delta_f H^\ominus / \text{kJ mol}^{-1}$	-658	-92.3

Calculate the standard enthalpy of formation, in kJ mol^{-1} , of $CHClF_2$

Give your answer to 3 significant figures.

[3 marks]

Enthalpy of formation _____ kJ mol^{-1}

6

END OF QUESTIONS



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ANSWER IN THE SPACES PROVIDED**



[illegible]

Question number	Additional page, if required. Write the question numbers in the left-hand margin.
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